

Makar Golotin

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EDUCATION

University of British Columbia, Vancouver, BC
Bachelor of Applied Science, Mechanical Engineering (Co-op)
Available May 2027 for a 4 month co-op placement

Sept 2024 – April 2029

EXPERIENCE

Engineering Co-op, Heidelberg Materials (Pipeline Aggregates)
Vancouver, BC

Apr 2026 – Present

- Ran RPAS drone survey flights over two active sites and processed the aerial maps in Propeller to track volume on roughly 100,000 tonnes of stockpiled material
- Led a vehicle segregation risk assessment on a shared haul road and designed a bypass route to separate heavy and light vehicles. Closed every hazard identified in the assessment
- Wrote emergency response plans and Safe Work Procedures under BC OHS regulation, including LOTOTO steps for site equipment
- Put the procedures on QR code posters at each site so crews could pull them up in the field
- Rebuilt the online safety orientation into a clearer sequence of modules, completed by all incoming workers
- Ran volume calculations for the 2026 pit mining plan (~1,000 tonne extraction) and discussed material placement with the site foreman
- Built inventory tracking systems for screen media and consumables
- Proposed structural improvements to strengthen site safety compliance
- Earned BC Mine Supervisor certification

Mechanical Project Lead (Battery), UBC ThunderBikes
Vancouver, BC

Sept 2024 – Present

- Led a 3-person sub-team designing a 540-cell (27s20p) lithium-ion battery pack for an electric motorcycle competing in MotoStudent 2027
- Designed a three-module pack, about 45 kg, that spreads out the series connections and makes the pack easier to service. Built a single 12 kg module first to test the layout
- Selected materials, PC-FR, ABS, Delrin and aluminum, based on thermal expansion and manufacturability analysis. Modeled assemblies and detailed parts in SolidWorks
- Managed a \$7,000 budget and coordinated mechanical and electrical work across the team, using version-controlled documentation

Shift Leader, The Patch Brewery
Maple Ridge, BC

Apr 2025 – Present

- Ran daily operations for a team generating ~\$40,000 in net sales per day: closing, financial reporting, scheduling and conflict resolution

PROJECTS

Claw Sigma, A Better Tomorrow Hackathon, UBC

Feb 2026

- Built a working 3-DOF Arduino-controlled robotic claw from concept to prototype in 7 hours using ~95% salvaged materials
- Combined 3D-printed fingers with reclaimed hardware machined in the student shop
- Won Most Ready-for-Market Prototype as the only team to deliver a functioning build instead of a concept pitch

Arduino-Controlled Autonomous Claw, UBC Applied Science

Feb – Apr 2025

- Designed and built a sheet metal autonomous claw with sensing and motor control on Arduino
- Earned the top performance result in the class section

TECHNICAL SKILLS

CAD & Analysis	SolidWorks, GD&T, DFM/DFA, FEA, Tolerance Analysis, Thermal Analysis
Fabrication	Mills & Lathes, 3D Printing, Waterjet & Laser Cutting, Soldering, Hand & Power Tools
Software	MATLAB, C, Arduino, Propeller (drone photogrammetry), Excel, MS Office Suite
Certifications	BC Mine Supervisor, RPAS Basic, Machine Shop Safety, WHMIS, Fall Protection, BC Driver's Licence
Languages	English (fluent), Russian (fluent), French (intermediate)